

ロボットの模倣学習における  
位置制約付き自律動作のためのポテンシャル場

Potential Fields for Position-Constrained Autonomous Motion  
in Robotic Imitation Learning

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**Abstract** – The use of bilateral teleoperation has enabled imitation learning with force control. However, when an obstacle appears on the motion path, imitation learning fails to avoid it due to the lack of an obstacle avoidance algorithm. In such cases, it becomes necessary to reacquire data and repeat the entire learning process whenever the environment changes. In this study, a potential field is applied to an environment with obstacles that were not present during the learning phase. A virtual force is applied to the robot's input during autonomous motion to alter its trajectory, thereby enabling the robot to avoid obstacles without requiring additional data. However, learning models with recursive structures may exhibit unexpected behaviors due to their non-Markovian nature. To address this issue, we explore the possibility of achieving adaptive behavior to potential fields by employing a hierarchical model in which the lower layers ensure Markov properties. Following imitation learning using bilateral control, we validate whether adaptive behavior to environmental changes is feasible by applying a potential field during autonomous behavior in pick-and-place tasks. Furthermore, we evaluate the effectiveness of the proposed hierarchical learning model by comparing it with a non-hierarchical model composed solely of long short-term memory (LSTM).

参考文献